

Activity 3

NAME: _____

Q.1. Solve the equation: $\frac{3}{7x-2} = \frac{9}{3x+1}$

Q.2.

31 Lawn mowing rates It takes a boy 90 minutes to mow the lawn, but his sister can mow it in 60 minutes. How long would it take them to mow the lawn if they worked together, using two lawn mowers?

Q.3. Complete the square: $4x^2 - 12x - 11$?

Q.4. Solve for x in terms of y : $4x^2 - 4xy + 1 - y^2 = 0$

Q.5.

69 Distance between airplanes An airplane flying north at 200 mi/hr passed over a point on the ground at 2:00 P.M. Another airplane at the same altitude passed over the point at 2:30 P.M., flying east at 400 mi/hr (see the figure).

- (a) If t denotes the time in hours after 2:30 P.M., express the distance d between the airplanes in terms of t .
- (b) At what time after 2:30 P.M. were the airplanes 500 miles apart?

Q.6. Write as a single number of the form $a + bi$ with a and b real: $\frac{-3-2i}{5+2i}$

Q.7. Find all real values of x and y satisfying the following equation: $(x - y) + 3i = 7 + yi$